

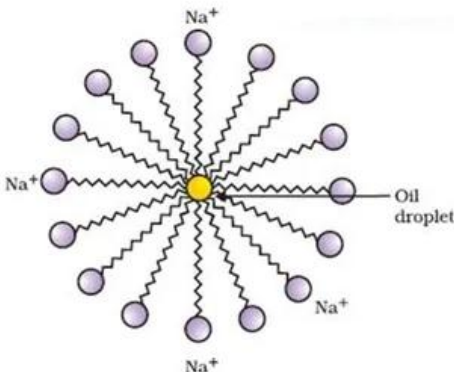
PRE-BOARD EXAMINATION -1 (2024 -25)
MARKING SCHEME
Class X Science (086)

Section-A		
1	A. Only III	1
2	C. Displacement reaction	1
3	B. Cooking of food	1
4	A. Yellow	1
5	B. a – (iii), b – (i), c – (iv), d – (ii)	1
6	D. sex is not genetically determined in snails.	1
7	A. C ₂ H ₅ OH	1
8	C. Setup R	1
9	B. Transpiration	1
10	B. Has low melting point	1
11	D. II and IV	1
12	D. flow of water in and out of the guard cells	1
13	B. A ray passing through the centre of curvature and incident obliquely.	1
14	B. 1 and 4	1
15	C. CFCs; Ozone	1
16	B. Enrichment of oxygen in the atmosphere.	1
17	D. A is false but R is true.	1
18	A. Both A and R are true, and R is the correct explanation of A.	1
19	C. A is true but R is false.	1
20	C. A is true but R is false	1

Section-B		
21	- To remove nitrogenous waste (0.5) - To remove excess water (0.5) - Urine is stored in urinary bladder (0.5) - Urine is released by ureter (0.5)	2
22	A. Exothermic/ Displacement reaction/Redox reaction. (0.5) Heat is evolved or a more reactive element displaces a less reactive element or aluminium reduces iron (II) oxide to iron. (0.5) B. Double displacement / Precipitation reaction (0.5) As there is an exchange of ions between reactants and products / yellow precipitate (of Lead iodide) is formed (0.5)	2
23	<u>Students to attempt either option A</u> A. (i) Salivary amylase – breaks down starch which is a complex molecule to sugar. (ii) Pepsin – Helps to digest proteins in stomach. (iii) Trypsin – It helps in digesting proteins to amino acids. (iv) Lipase – Breaking down of emulsified fats to fatty acids and glycerol. (0.5 x 4) OR B. a) Water (0.5) b) (i) Conversion of light energy to chemical energy and splitting of water molecules into hydrogen and oxygen. (ii) Reduction of carbon dioxide into carbohydrates (0.5 x 2) c) Light energy to chemical energy (0.5)	2
24	(i). Decrease object distance (ii). Increase focal length of the lens (1x2)	2

25	<u>Student to attempt either A or B.</u> A. (1) i. Both have same reading ii. Reason- Both are connected in series and in series current remains the same. (1) OR B. $P = V \times I$ $I = P/V = 1000/220 = 4.54 \text{ A}$ (1) We will be using fuse B with is rated as 5A. This is because it will be able to sustain the current (4.54 A) passing through it. Whereas fuse A will melt and break the circuit as the current exceeds its rating. (1)	2
26	i. Producers and Secondary consumers (carnivores) respectively ii. Producers synthesize their own food Carnivores feed on herbivores/ other animals	1+1
Section-C		
27	A. $R \propto l$ $R \propto l/A$ $R = \rho l/A$. (1.5) B. $R = \rho l/A$ $\rho = RA/l$ for $A = 1\text{m}^2$ and $l = 1\text{m}$, we have $\rho = R$ Hence resistivity is the resistance offered by a wire of length 1 m having a cross sectional area of 1 m^2 . (1.5)	3
28	<u>Students to attempt either A or B.</u> A. (i) Dry HCl (1) (ii) HCl gas does not ionize in dry conditions. Hence it does not act as an acid, so it can not change blue litmus to red. (1) (iii) To absorb moisture (1) OR B. (i) $X = \text{NaHCO}_3$; $Z = \text{Na}_2\text{CO}_3$ (0.5+0.5) (ii) Decomposition reaction (0.5) (iii) Solution A (0.5) (iv) Increasing order of H^+ ions $C < B < A$ (1)	3
29	• Cranium and Cerebrospinal fluid (1) • i. Cerebellum, voluntary (1) • ii Medulla Oblongata, involuntary/ reflex (1)	3

30	(Any letter which clearly indicates dominant and recessive ears, example, L or E or any other) A. $LL \times ll$ $F_1 = Ll$ $Ll \times Ll$ $1(LL):2(Ll):1(ll)$ (2) B. No change in ratio/the ratio of F_2 generation will still be $1(LL):2(Ll):1(ll)$/ ratio will be the same. (1) As the cross is still between a pure dominant and recessive allele/ genes/ traits/characters /as shown in the cross above. (1)	3
31	i. No, because convex lens is a converging lens and it will reduce the focal length further and the image will not form on the retina. ii. $f = 1/P = 1/(-2) = -0.5 \text{ m}$ iii. Given $u = -10 \text{ cm}$; $P = -2 \text{ D}$ $f = -0.5 \text{ m} = -50 \text{ cm}$ $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ $v = \frac{uf}{u+f}$ $v = \frac{(-10)(-50)}{-10-50}$ $v = -8.33 \text{ cm}$	$1 \times 3 = 3$
32	i. Metal X is Zinc (1) ii. Roasting + equation $2ZnS + 3O_2 \xrightarrow{\text{Heat}} 2ZnO + 2SO_2$ (1/2 + 1/2) iii Reduction + equation $2ZnO + C \xrightarrow{\text{Heat}} 2Zn + CO$ (1/2 + 1/2)	3
33	A. Magnetic field strength is inversely proportional to the distance from the current carrying wire. Hence when the compass is moved away from the current carrying wire, the magnetic effect was less on it and hence the deflection was less. (1) B. Magnetic field strength is directly proportional to the current in the wire. So, increasing the current in the circuit will produce a greater deflection in the compass needle. (1) C. Direction of magnetic field lines is given by Right hand thumb rule. Reversing the polarity of the battery will thus reverse the direction of magnetic field lines. (1)	3
Section-D		
34	<u>Student to attempt either option A or B.</u> A. i). Compound A = Ethanol B = Ethene C = Ethane (1.5) ii). Correct equation $\text{CH}_3\text{CH}_2\text{OH} \xrightarrow[170^\circ\text{C}/443\text{K}]{\text{conc. H}_2\text{SO}_4} \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$ (1) iii. CO_2, H_2O and heat are produced (1) iv. Production of vanaspati ghee from vegetable oils (1) v. Sodium ethoxide (0.5)	5

	OR B. (i) Soap and Detergent respectively (1) (ii)  (1) iii. A = soap forms scum . (0.5 + 1) The calcium and magnesium salts present in hard water are precipitated with carboxylic acids to form scum. (The Calcium and Magnesium salts of sulphonic acid are soluble in water. 'B' is a more effective cleansing agent in presence of Ca and Mg salts.) (iii) $\text{CH}_3\text{COOCH}_3 + \text{NaOH} \longrightarrow \text{CH}_3\text{COONa} + \text{CH}_3\text{OH}$ OR $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{NaOH} \longrightarrow \text{CH}_3\text{COONa} + \text{C}_2\text{H}_5\text{OH}$ The process is saponification. (1 + 0.5)	
35	<u>Students to attempt either option A or B.</u> A. (i) Correct examples (0.5+0.5) (ii) More crops in same time interval, genetically identical, flower fruit faster, no need to depend on pollinators. (1+1) (iii) 1. Cross pollination, the pollen from anther will be transferred the stigma of another bryophyllum plant using agents like wind, water, or insects etc. (0.5+0.5) 2. There would be minor changes/some variation during the process of copying of the DNA. (1) OR B. (i) a. Zygote divides to form embryo which gets implanted in the uterus wall to develop into a foetus. (1) b. Menstruation - Unfertilized egg comes out through vaginal passage along with blood lining and mucus. (1) (ii) Uterus; has thick lining with rich supply of blood to nourish the embryo. (1) (iii) Transfer/ Exchange of nutrients, gases and waste occurs through a special tissue called placenta. It has villi like projections to increase surface area for transfer of materials. (2)	5

36	<u>Students to attempt either option A or B.</u> A. i. Ohm's Law is the law, which states that the electric current (I) flowing through a conductor is directly proportional to the voltage (V) applied across it and inversely proportional to the resistance (R) of the conductor. Mathematically, it can be represented as: (1) $V \propto I = IR \quad (1)$ (iii) Let the energy consumed by the fans be E_B and the energy consumed by the electric kettle be E_K $E_B = P \times t = 200 \times 5 / 1000 = 1 \text{ kWh} \quad (1)$ $E_K = P \times t = 1000 \times 2 / 1000 = 2 \text{ kWh} \quad (1)$ Total energy consumed = $E_B + E_K = 3 \text{ kWh}$ Total energy consumed in 60 days = 90 kWh Total cost = $90 \times \text{Rs } 7.5 = \text{Rs } 675 \quad (1)$ OR B. (i) Joule's Law of Heating states that the amount of heat produced in a conductor is directly proportional to the square of the electric current passing through it, the resistance of the conductor, and the time for which the current flows. Mathematically, it can be expressed as $H = I^2 R t$ • H is the heat produced (in joules), • I is the electric current (in amperes), • R is the resistance of the conductor (in ohms), • t is the time for which the current flows (in seconds). (ii) Let the equivalent resistance in series be denoted by R_s and that for parallel be denoted by R_p. Total voltage of the circuit is given by V in both cases and the time is denoted by t. • $R_s = 2 + 4 = 6 \Omega$ • $\frac{1}{R_p} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}, R_p = \frac{4}{3}$ • $H_s = \frac{V^2}{R_s} t, H_p = \frac{V^2}{R_p} t$ • $\frac{H_s}{H_p} = \frac{R_p}{R_s} = \frac{2}{9}$ (2+3)	5
Section – E		
37	A. $\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \text{C} - \text{H} \\ \\ \text{CH}_3 \end{array}$ (1)	4

	B $\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_2 = \text{C} \\ \\ \text{CH}_3 \end{array}$ (1) <u>Student to attempt either subpart C OR D</u> $\text{H}_3\text{C} - \text{CH}_2 - \text{CH} = \text{CH}_2 \quad \text{H}_3\text{C} - \text{CH} = \text{CH} - \text{CH}_3$ (2) OR D (i) one or two (1) (no marks if both 1 or 2 not given) (ii) six (1)	
38	<u>Students to attempt either subpart A or B.</u> A. ● Phototropism (0.5) ● When light is coming from one side of the plant, auxin diffuses to the shady side of the shoot causing the cells on that side to elongate (1) ● This concentration of auxin causes bending of shoot towards light. (0.5) OR B. ● Leaves of ‘Touch me not’ plant respond to the stimulus by showing growth independent movement. ● These plants use electrical–chemical means to convey the information from cell to cell. ● Movement happens at a point different from the point of touch. ● Plant cells change shape by changing the amount of water in them, resulting in swelling or shrinking, and therefore in changing shape. (0.5 x 4 =2) C. Growth of pollen tubes towards the ovule is an example of chemotropism whereas bending of shoots towards sunlight is an example of phototropism. (1)	4
	D. i) Although both plants and animals show electrical–chemical means to convey the information from cell to cell but unlike nerve cells in animals there is no specialized tissue in plants for conduction of information. (0.5) ii) In animal cells, change in shape occurs because of the specialized proteins found in muscle cells; plant cells change shape by changing the amount of water in them. (0.5)	

39	A. Virtual, erect and magnified (1) B. Less than 6 cm from the optical centre (1) <u>Student to attempt either subpart C or D.</u> C. Correct ray diagram (2) OR D. Same size real and inverted image is formed by a convex lens when the object is placed at $2f_1$ Therefore, $u = v = 2f$ $2f + 2f = 60\text{cm}$ $4f = 60\text{ cm}$ $F = 15\text{ cm}$	4
----	--	---
